PS : 1 / 2010
PERFORMANCE AND CONSTRUCTION OF ELECTRIC FANS AND REGULATORS

Clause	Requirement – Test	Result - Remark	Verdict
9.3.2	Fans shall be tested at rated frequency, if marked. For a fan with a range of frequencies, the tests shall be made at the frequency which gives the most unfavourable results. For a fan not marked with rated frequency the tests shall be made either at 50Hz or 60Hz whichever is more unfavourable.	50 Hz	P
9.3.3	***Limits of voltage variation*** The variation in the voltage shall not exceed ± 1% of the test voltage during air performance tests, while taking the current and wattage readings during these tests, however, the voltage shall be the test voltage.	-	P
9.4	***Test for Air Performance (Air Delivery Test)*** The test shall be carried out at an ambient temperature of 20 ± 5°C The air delivery measured in accordance with test method described for determining the air performances of a particular type of fan shall not be less than the permissible value as given in Appendix – A of PS-1/2010.	Measured air delivery = 246.2 m^3/min Standard requirement: Air delivery= 250 m^3/min Deviation = –1.52 % Permissible deviation = –10 %	P
9.5	***Measurement of speed of the fan*** The measured value of fan speed shall not be grater than the permissible value as given in the Appendix – A of PS-1/2010.	Measured speed = 328 rpm Standard requirement: Maximum speed ≤ 330 rpm	P
9.6	***Measurement of power factor (for a.c fans only) and power input*** Power factor shall not less than the 0.9 permissible value as given in the Appendix – A of PS-1/2010.	Measured power factor = 0.98 Standard requirement: Power factor = 0.9	P
	The measured power input shall not be grater than the permissible value as given in the Appendix – A of PS-1/2010.	Measured power input = 54.5 W Standard requirement: Power input = 80 W Deviation = –31.88 % Permissible deviation = +10 %	P
-	***Service Value*** The service value shall not be less than the minimum value as per Appendix – A of PS-1/2010.	Measured service value = 4.518 m^3/min/w Standard requirement: Min. service value ≥ 3.12 m^3/min/w 3-Star NEECA requirement: Min. service value ≥ 3.45 m^3/min/w	P

Note:
1. The speed reduction test was not performed as the speed regulator was not provided by the manufacturer.
2. The rated values of air delivery and power input are given as per Appendix – A of PS-1/2010.
3. Air Velocity Curve of the tested sample of ceiling fan is attached as Annexure –1.
4. The Photographs of the tested product is attached as Annexure –2.

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